A Kernel and Resolvent Certificate for Zero-Drift Skew Brownian Motion

Route-A source-local certificate HCS-C266

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Abstract

We freeze symmetric local time and derive the conservative transition kernel, generator interface, and complete resolvent of zero-drift skew Brownian motion. The statement retains ordinary Brownian motion and the two reflected endpoint faces. It uses no target arithmetic or spectral data.

1 Frozen process

Let B be standard Brownian motion and $L^0(X)$ symmetric semimartingale local time. For $0 \le p \le 1$, put $\theta = 2p - 1$ and freeze

$$X_t = x + B_t + \theta L_t^0(X). \tag{1}$$

Harrison and Shepp proved that (1) has a unique strong solution exactly for $|\theta| \le 1$ and identified it by assigning independent excursion signs, positive with probability p [1]. This fixes the normalization: a right-local-time equation would have a different coefficient.

Write $\phi_t(z) = (2\pi t)^{-1/2} e^{-z^2/(2t)}$.

Theorem 1 (kernel and interface)**.** *For $t > 0$, the Lebesgue transition density is*

$$q_p(t; x, y) = \phi_t(y - x) + (2p - 1)\,\mathrm{sgn}(y)\phi_t(|x| + |y|). \tag{2}$$

It is conservative and satisfies Chapman–Kolmogorov. On a Feller generator core, $Af = f''/2$ away from zero and

$$pf'(0+) = (1 - p)f'(0-). \tag{3}$$

Proof. In each open quadrant (2) is a sum of Gaussian images and solves the heat equation. As a function of x it is continuous at zero; its one-sided derivatives satisfy (3). Direct half-line Gaussian integration gives total mass one and positivity. Splitting the convolution at zero and applying the Gaussian semigroup identity gives Chapman–Kolmogorov. Thus (2) is the transition kernel of the excursion-sign solution of (1), and the backward interface condition is (3). □

The terminal-coordinate limits are generally unequal: $q_p(t; x, 0+)/q_p(t; x, 0-) = p/(1 - p)$. Thus (2) is not a continuous Lebesgue density unless $p = 1/2$; this is a feature, not a normalization defect.

Theorem 2 (resolvent)**.** *For $\lambda > 0$ and $k = \sqrt{2\lambda}$,*

$$r_\lambda(x, y) = \int_0^\infty e^{-\lambda t} q_p(t; x, y)\,dt = \frac{e^{-k|x-y|} + (2p - 1)\,\mathrm{sgn}(y)e^{-k(|x|+|y|)}}{k}. \tag{4}$$

Proof. Apply $\int_0^\infty e^{-\lambda t}\phi_t(z)\,dt = e^{-k|z|}/k$ to the two terms of (2). The result also solves $(\lambda - A)r = 0$ off $x = y, 0$, obeys (3), and has the unit derivative jump at the source point. □

At $p = 1/2$ the image term vanishes. At $p = 1$ or 0, after the first interface hit every excursion has one sign, giving the corresponding one-sided reflected face. We make no claim for drift, unequal diffusivities, or sticky interfaces.

References

[1] J. M. Harrison and L. A. Shepp, *On skew Brownian motion*, Ann. Probab. **9** (1981), 309–313, doi:10.1214/aop/1176994472.